

Example

$$p_r(r) = \begin{cases} \frac{2r}{(L-1)^2} & \text{for } 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

From Eq. (3.3-4),

$$s = T(r) = (L-1) \int_0^r p_r(w) dw = \frac{2}{L-1} \int_0^r w dw = \frac{r^2}{L-1}$$

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$$s = T(r) = (L-1) \int_0^r p_r(w) dw = \frac{2}{L-1} \int_0^r w dw = \frac{r^2}{L-1}$$

$$\begin{aligned} p_s(s) &= p_r(r) \left| \frac{dr}{ds} \right| = \frac{2r}{(L-1)^2} \left| \left[\frac{ds}{dr} \right]^{-1} \right| \\ &= \frac{2r}{(L-1)^2} \left| \left[\frac{d}{dr} \frac{r^2}{L-1} \right]^{-1} \right| \\ &= \frac{2r}{(L-1)^2} \left| \frac{(L-1)}{2r} \right| = \frac{1}{L-1} \end{aligned}$$

- For discreet values,

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$$p_r(r_k) = \frac{n_k}{MN} \quad k = 0, 1, 2, \dots, L - 1 \quad (3.3-7)$$

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The discrete form of the transformation in Eq. (3.3-4) is

$$\begin{aligned} s_k = T(r_k) &= (L - 1) \sum_{j=0}^k p_r(r_j) \\ &= \frac{(L - 1)}{MN} \sum_{j=0}^k n_j \quad k = 0, 1, 2, \dots, L - 1 \end{aligned} \quad (3.3-8)$$

- For discrete values,

$$p_r(r_k) = \frac{n_k}{MN} \quad k = 0, 1, 2, \dots, L - 1 \quad (3.3-7)$$

The discrete form of the transformation in Eq. (3.3-4) is

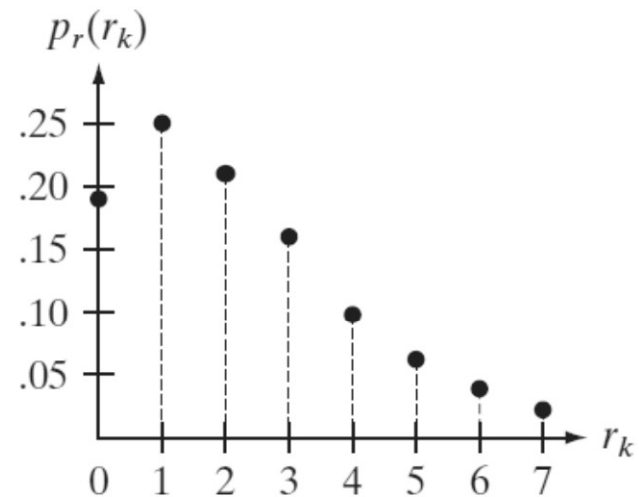
$$\begin{aligned} s_k = T(r_k) &= (L - 1) \sum_{j=0}^k p_r(r_j) \\ &= \frac{(L - 1)}{MN} \sum_{j=0}^k n_j \quad k = 0, 1, 2, \dots, L - 1 \end{aligned} \quad (3.3-8)$$

Thus, a processed (output) image is obtained by mapping each pixel in the input image with intensity r_k into a corresponding pixel with level s_k in the output image, using Eq. (3.3-8). The transformation (mapping) $T(r_k)$ in this equation is called a *histogram equalization* or *histogram linearization* transformation. It is not difficult to show (Problem 3.10) that this transformation satisfies conditions (a) and (b) stated previously in this section.

Histogram Equalization Numerical

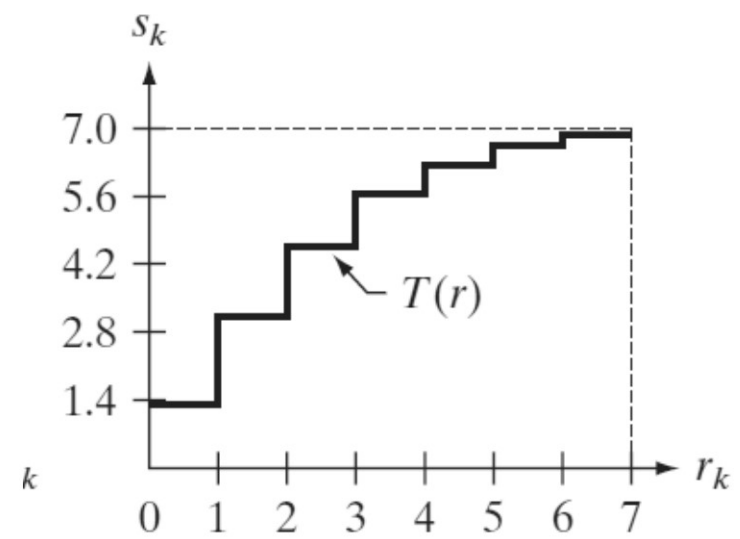
Suppose that a 3-bit image ($L = 8$) of size 64×64 pixels ($MN = 4096$) has the intensity distribution shown in Table 3.1, where the intensity levels are integers in the range $[0, L - 1] = [0, 7]$.

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

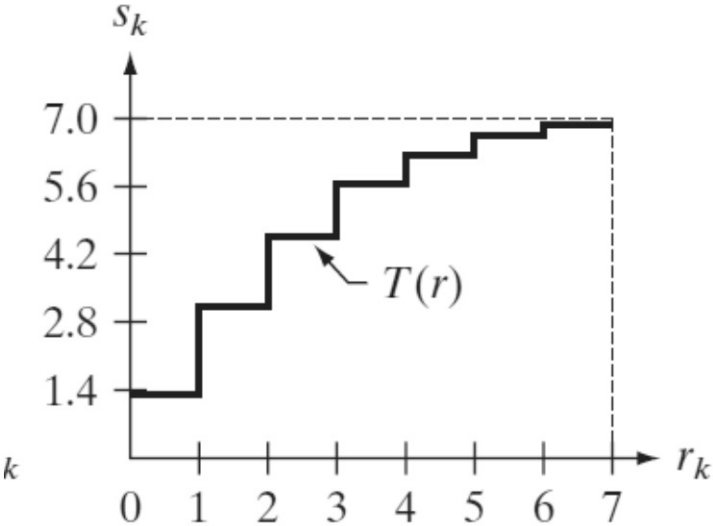


$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

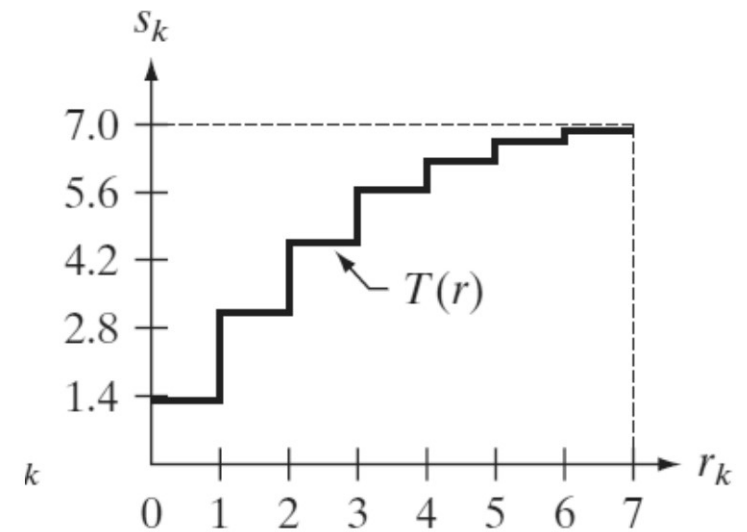


$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$



r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
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$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

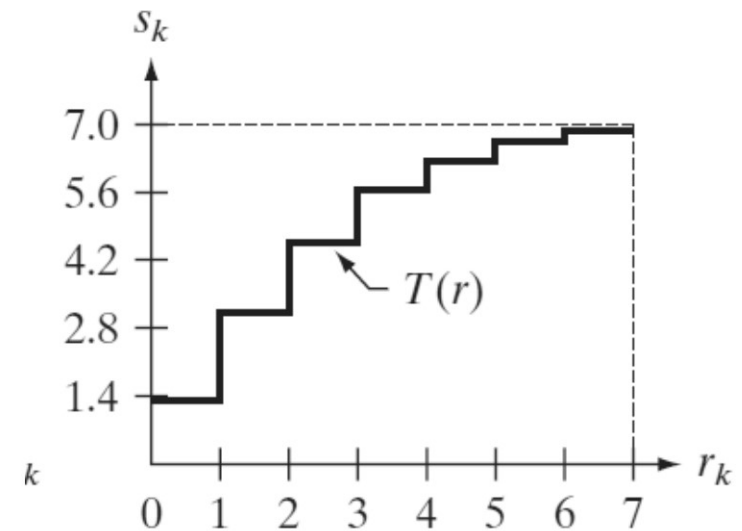
$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$



r_k	n_k	$p_r(r_k) = n_k/MN$
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$r_7 = 7$	81	0.02

$$CDF = \sum_{j=0}^k p_r(r_j)$$

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

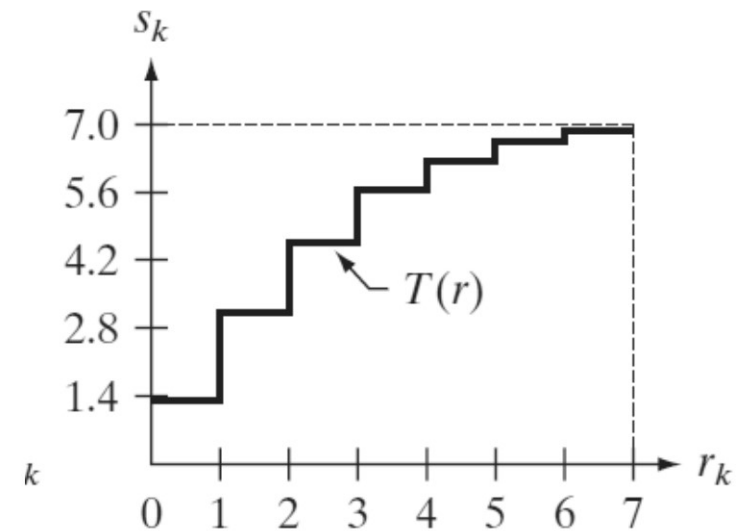


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$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$\text{CDF} = \sum_{j=0}^k p_r(r_j)$$

0.19
 $0.19 + 0.25 = 0.44$

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

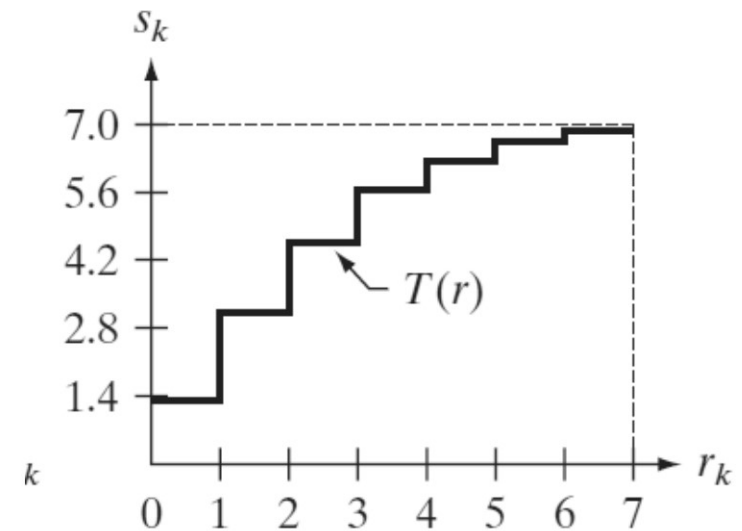


r_k	n_k	$p_r(r_k) = n_k/MN$
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$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
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$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$CDF = \sum_{j=0}^k P_r(r_j)$$

$$0.19 \quad 0.19 + 0.25 = 0.44 \quad \sum_{j=0}^1 P_r(r_j) = P_0(0) + P_1(1)$$

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

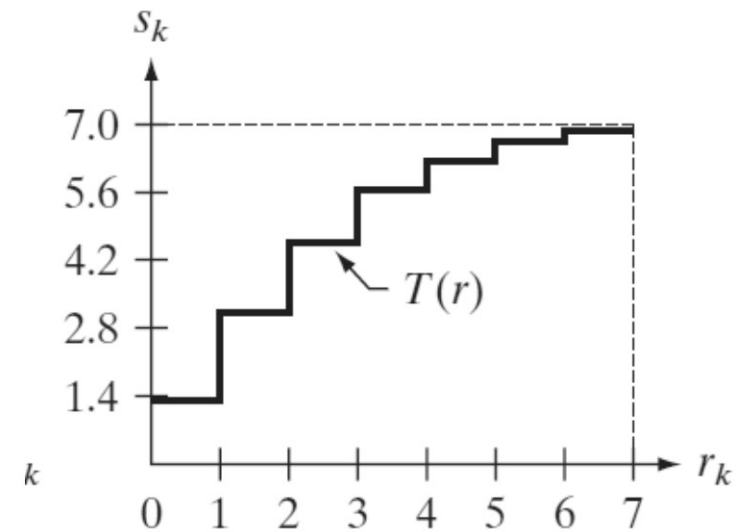


r_k	n_k	$p_r(r_k) = n_k/MN$
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$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$CDF = \sum_{j=0}^k p_r(r_j)$$

0.19
 $0.19 + 0.25 = 0.44$

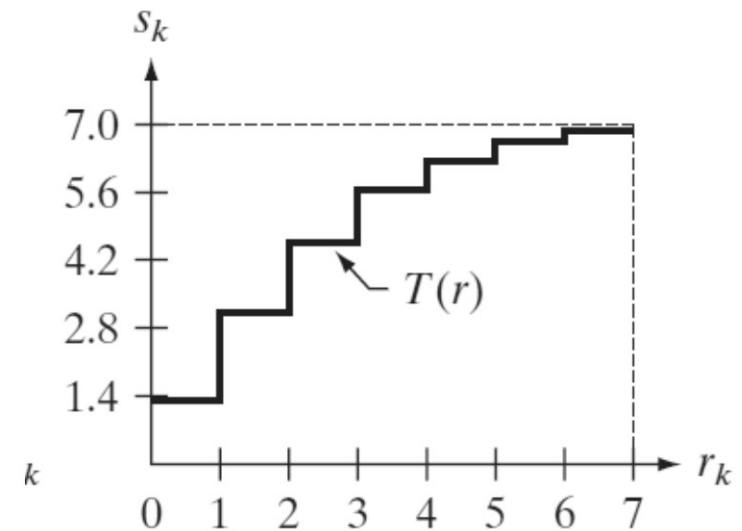
$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$



r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$CDF = \sum_{j=0}^k p_r(r_j)$
 0.19
 $0.19 + 0.25 = 0.44$
 0.65
 0.81

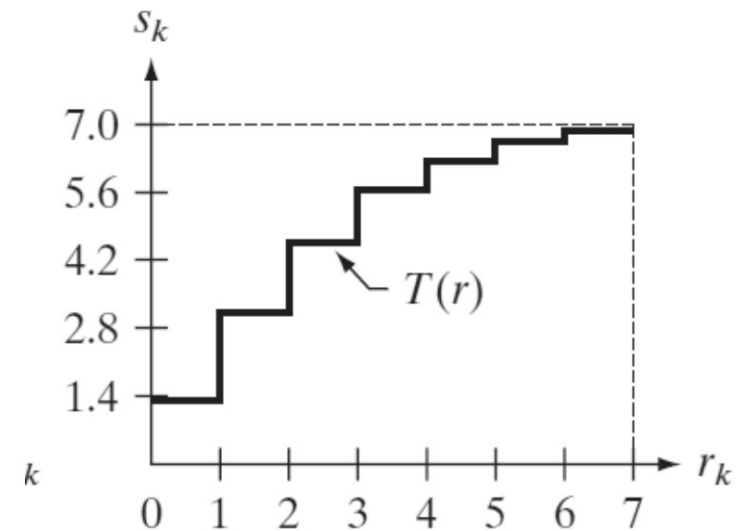
$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$



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$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$CDF = \sum_{j=0}^k p_r(r_j)$
 0.19
 $0.19 + 0.25 = 0.44$
 0.65
 0.81
 0.89
 0.95
 0.98
 1

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

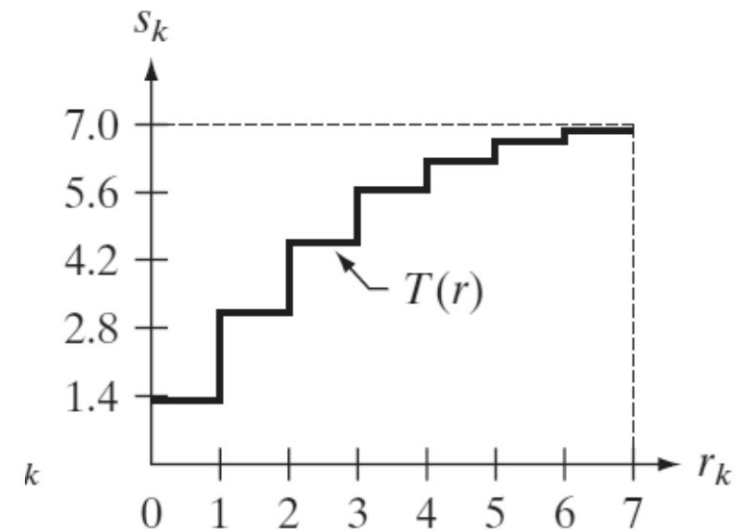


r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
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$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$CDF = \sum_{j=0}^k p_r(r_j)$
 0.19
 $0.19 + 0.25 = 0.44$
 0.65
 0.81
 0.89
 0.95
 0.98
 1

$$s_k = (L - 1) \times CDF$$

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

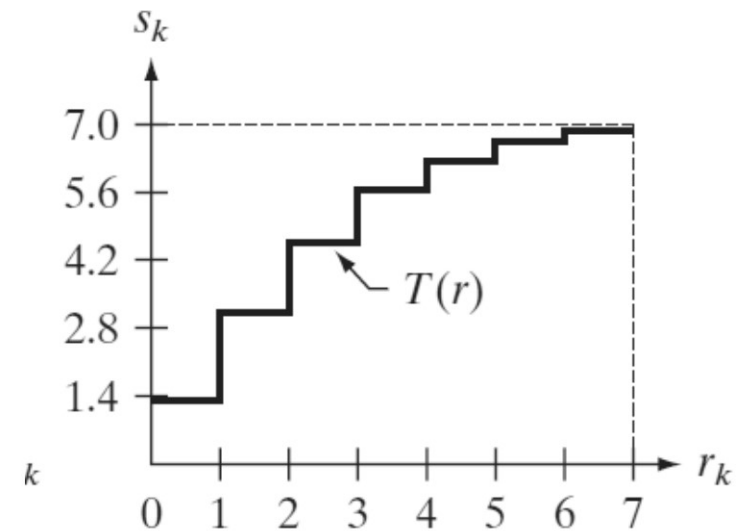


r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$\begin{aligned} \text{CDF} &= \sum_{j=0}^k p_r(r_j) \\ &0.19 \\ &0.19 + 0.25 = 0.44 \\ &0.65 \\ &0.81 \\ &0.89 \\ &0.95 \\ &0.98 \\ &1 \end{aligned}$$

$$\begin{aligned} s_k &= \overbrace{(L-1)}^7 \times \text{CDF} \\ 7 \times 0.19 &= 1.33 \\ 7 \times 0.44 &= 3.08 \end{aligned}$$

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$



r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

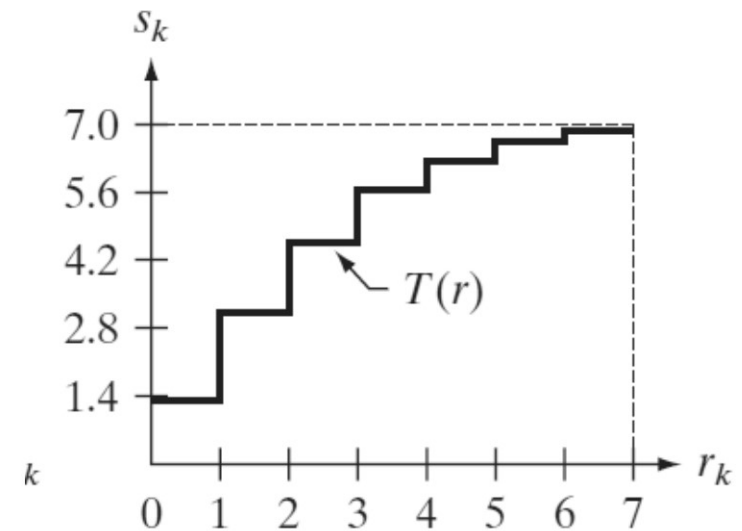
$$\text{CDF} = \sum_{j=0}^k p_r(r_j)$$

0.19
 0.19 + 0.25 = 0.44
 0.65
 0.81
 0.89
 0.95
 0.98
 1

$$s_k = (L-1) \times \text{CDF}$$

$7 \times 0.19 = 1.33$
 $7 \times 0.44 = 3.08$
 7×0.65
 7×0.81
 7×0.89
 7×0.95
 7×0.98
 7×1

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$



r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

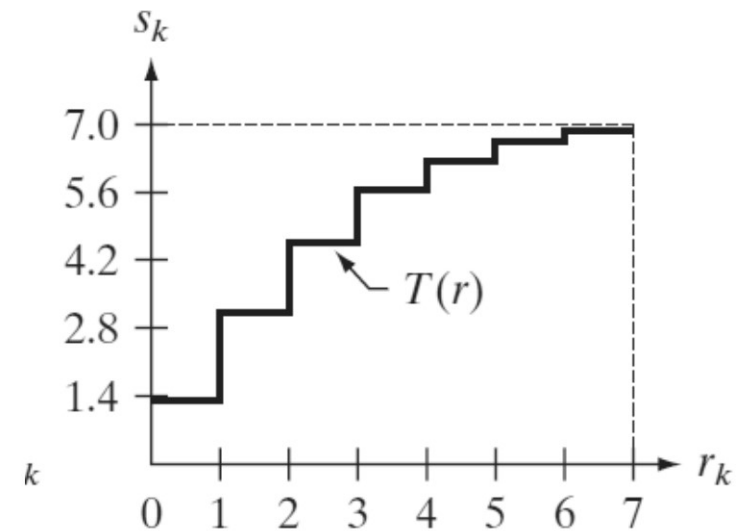
$$CDF = \sum_{j=0}^k p_r(r_j)$$

0.19
 0.19 + 0.25 = 0.44
 0.65
 0.81
 0.89
 0.95
 0.98
 1

$$S_k = (L-1) \times CDF$$

$7 \times 0.19 = 1.33$
 $7 \times 0.44 = 3.08$
 $7 \times 0.65 = 4.55$
 $7 \times 0.81 = 5.67$
 $7 \times 0.89 = 6.23$
 $7 \times 0.95 = 6.65$
 $7 \times 0.98 = 6.86$
 $7 \times 1 = 7$

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$



r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$CDF = \sum_{j=0}^k p_r(r_j)$
 0.19
 $0.19 + 0.25 = 0.44$
 0.65
 0.81
 0.89
 0.95
 0.98
 1

$S_k = (L-1) \times CDF$
 $7 \times 0.19 = 1.33 \rightarrow 1$
 $7 \times 0.44 = 3.08 \rightarrow 3$
 $7 \times 0.65 = 4.55 \rightarrow 5$
 $7 \times 0.81 = 5.67 \rightarrow 6$
 $7 \times 0.89 = 6.23 \rightarrow 6$
 $7 \times 0.95 = 6.65 \rightarrow 7$
 $7 \times 0.98 = 6.86 \rightarrow 7$
 $7 \times 1 = 7 \rightarrow 7$

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

$$s_0 = T(r_0) = 7 \sum_{j=0}^0 p_r(r_j) = 7p_r(r_0) = 1.33$$

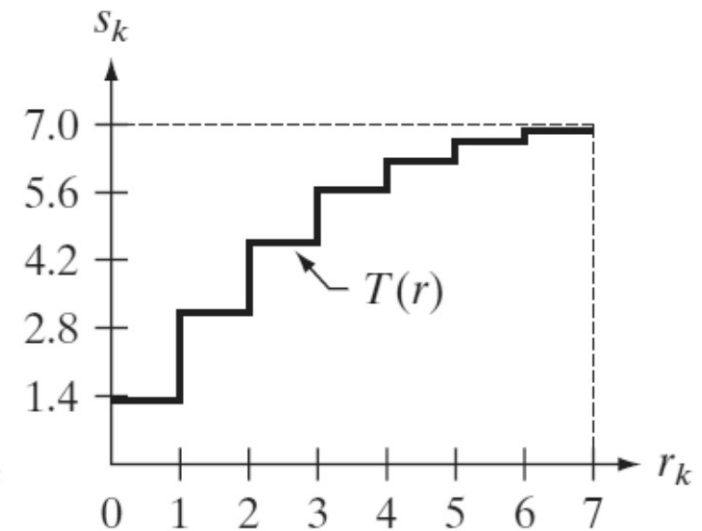
Similarly,

$$s_1 = T(r_1) = 7 \sum_{j=0}^1 p_r(r_j) = 7p_r(r_0) + 7p_r(r_1) = 3.08$$

and $s_2 = 4.55, s_3 = 5.67, s_4 = 6.23, s_5 = 6.65, s_6 = 6.86, s_7 = 7.00$.

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$\begin{aligned} \text{CDF} &= \sum_{j=0}^k p_r(r_j) \\ 0.19 \\ 0.19 + 0.25 &= 0.44 \\ 0.65 \\ 0.81 \\ 0.89 \\ 0.95 \\ 0.98 \\ 1 \end{aligned}$$



$$\begin{aligned} S_k &= (L-1) \times \text{CDF} \\ 7 \times 0.19 &= 1.33 \rightarrow 1 \\ 7 \times 0.44 &= 3.08 \rightarrow 3 \\ 7 \times 0.65 &= 4.55 \rightarrow 5 \\ 7 \times 0.81 &= 5.67 \rightarrow 6 \\ 7 \times 0.89 &= 6.23 \rightarrow 6 \\ 7 \times 0.95 &= 6.65 \rightarrow 7 \\ 7 \times 0.98 &= 6.86 \rightarrow 7 \\ 7 \times 1 &= 7 \rightarrow 7 \end{aligned}$$

At this point, the s values still have fractions because they were generated by summing probability values, so we round them to the nearest integer:

$$s_0 = 1.33 \rightarrow 1 \qquad s_4 = 6.23 \rightarrow 6$$

$$s_1 = 3.08 \rightarrow 3 \qquad s_5 = 6.65 \rightarrow 7$$

$$s_2 = 4.55 \rightarrow 5 \qquad s_6 = 6.86 \rightarrow 7$$

$$s_3 = 5.67 \rightarrow 6 \qquad s_7 = 7.00 \rightarrow 7$$

At this point, the s values still have fractions because they were generated by summing probability values, so we round them to the nearest integer:

$$s_0 = 1.33 \rightarrow 1 \qquad s_4 = 6.23 \rightarrow 6$$

$$s_1 = 3.08 \rightarrow 3 \qquad s_5 = 6.65 \rightarrow 7$$

$$s_2 = 4.55 \rightarrow 5 \qquad s_6 = 6.86 \rightarrow 7$$

$$s_3 = 5.67 \rightarrow 6 \qquad s_7 = 7.00 \rightarrow 7$$

r_k	n_k	$p_r(r_k) = n_k/MN$
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$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

At this point, the s values still have fractions because they were generated by summing probability values, so we round them to the nearest integer:

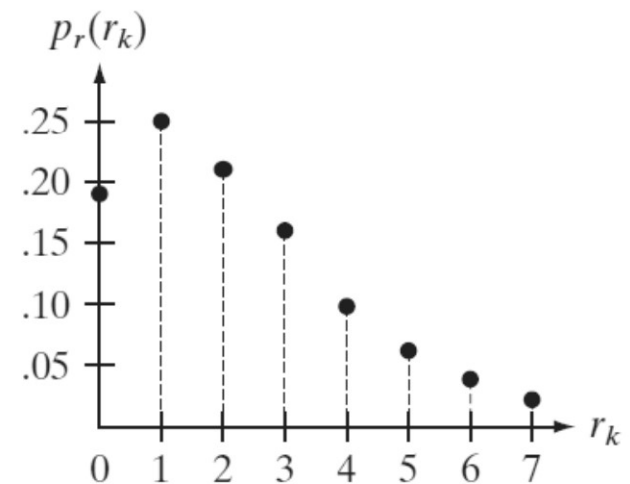
$$s_0 = 1.33 \rightarrow 1 \qquad s_4 = 6.23 \rightarrow 6$$

$$s_1 = 3.08 \rightarrow 3 \qquad s_5 = 6.65 \rightarrow 7$$

$$s_2 = 4.55 \rightarrow 5 \qquad s_6 = 6.86 \rightarrow 7$$

$$s_3 = 5.67 \rightarrow 6 \qquad s_7 = 7.00 \rightarrow 7$$

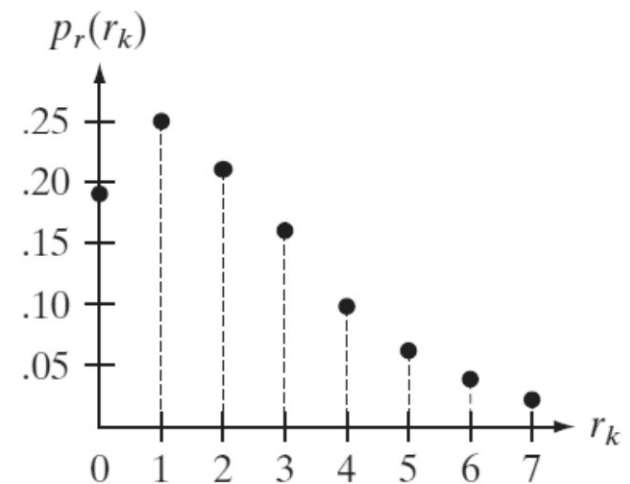
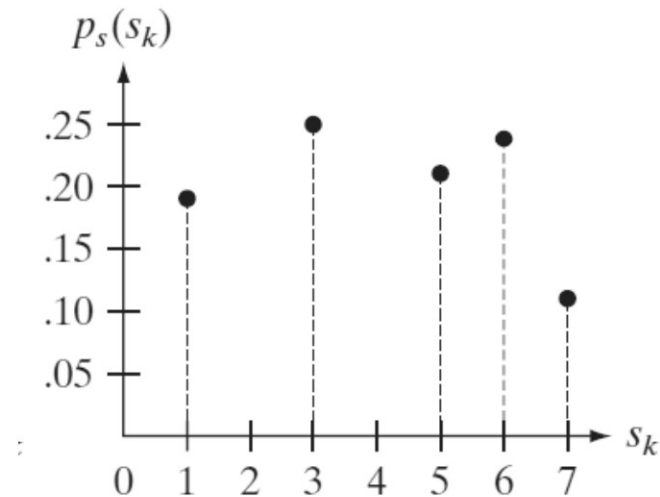
r_k	n_k	$p_r(r_k) = n_k/MN$
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$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02



At this point, the s values still have fractions because they were generated by summing probability values, so we round them to the nearest integer:

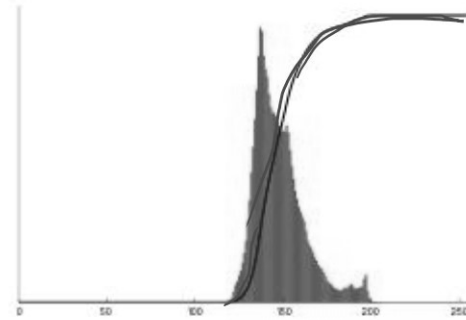
$$\begin{array}{ll} s_0 = 1.33 \rightarrow 1 & s_4 = 6.23 \rightarrow 6 \\ s_1 = 3.08 \rightarrow 3 & s_5 = 6.65 \rightarrow 7 \\ s_2 = 4.55 \rightarrow 5 & s_6 = 6.86 \rightarrow 7 \\ s_3 = 5.67 \rightarrow 6 & s_7 = 7.00 \rightarrow 7 \end{array}$$

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02





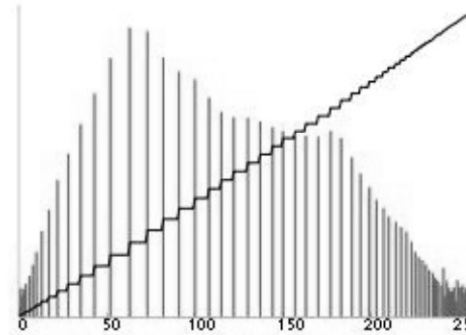
Before Histogram Equalization



Corresponding histogram (red) and cumulative histogram (black)



After Histogram Equalization



Corresponding histogram (red) and cumulative histogram (black)



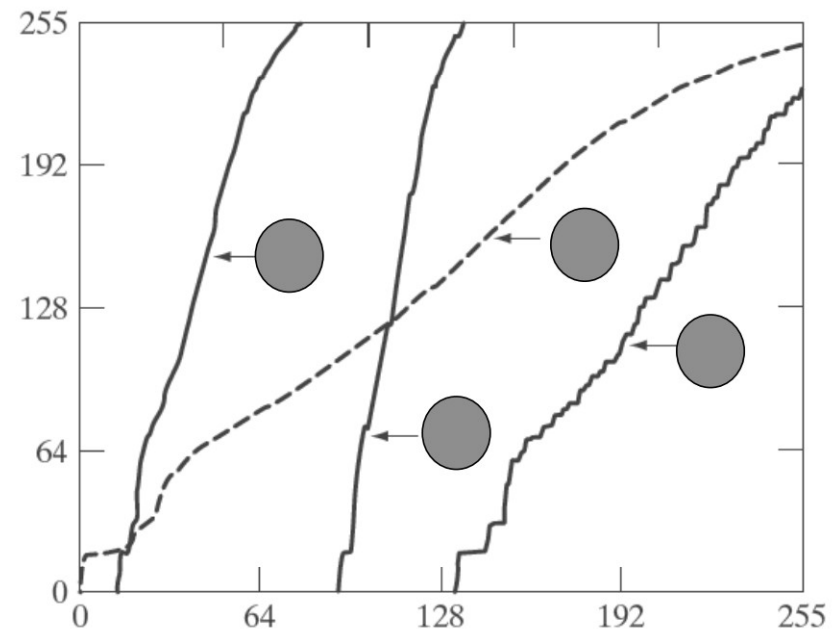


FIGURE 3.21
Transformation
functions for
histogram
equalization.
Transformations
(1) through (4)
were obtained from
the histograms of
the images (from
top to bottom) in
the left column of
Fig. 3.20 using
Eq. (3.3-8).

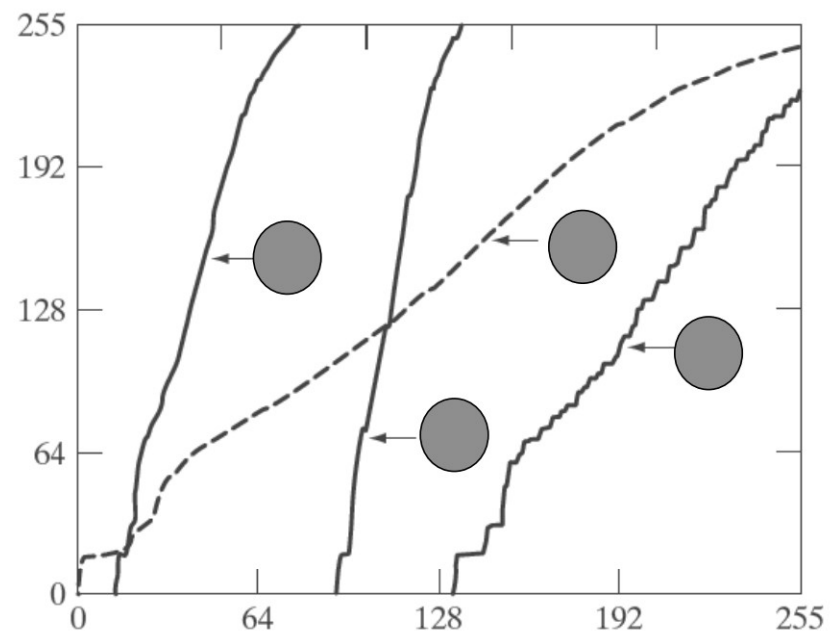
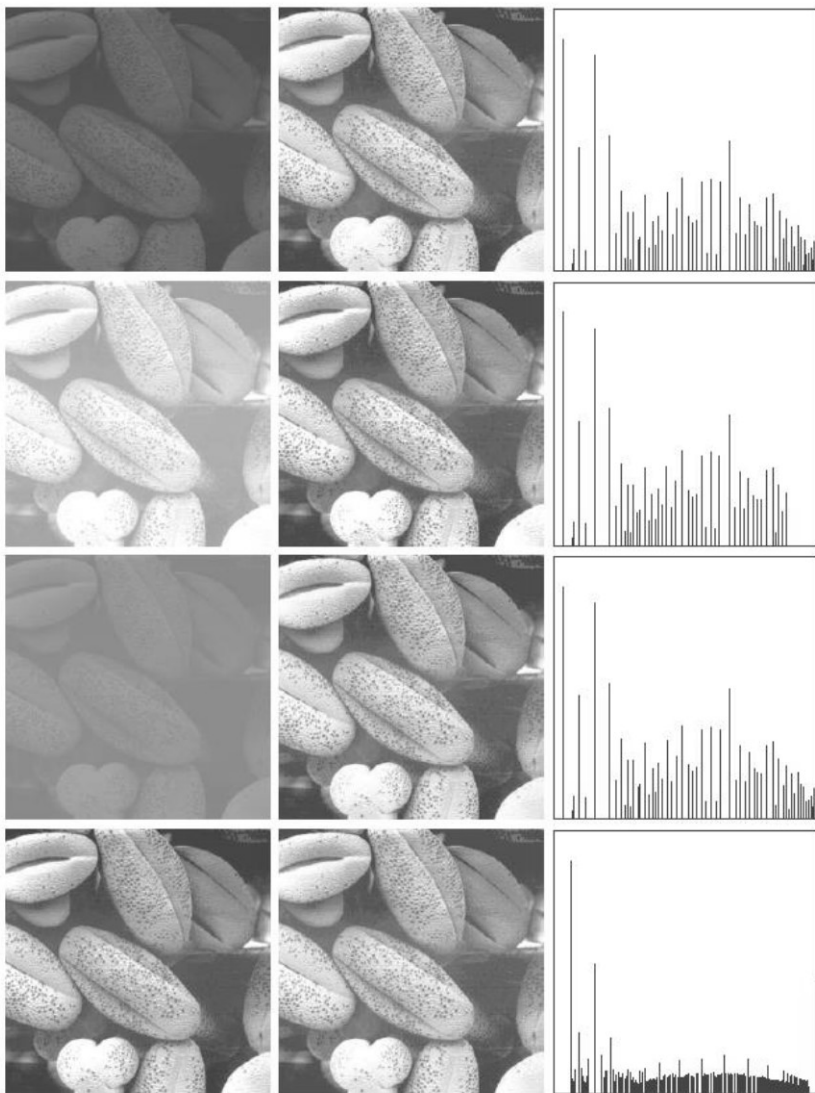


FIGURE 3.21 Transformation functions for histogram equalization. Transformations (1) through (4) were obtained from the histograms of the images (from top to bottom) in the left column of Fig. 3.20 using Eq. (3.3-8).

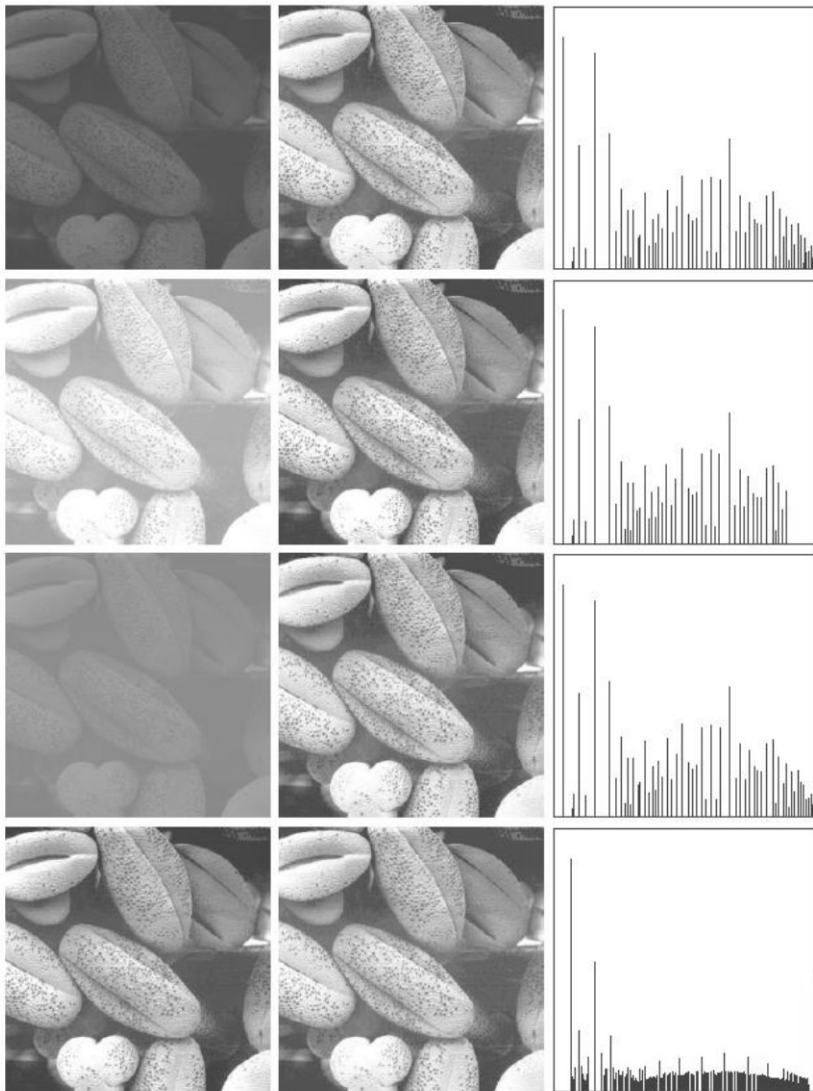


FIGURE 3.20 Left column: images from Fig. 3.16. Center column: corresponding histogram-equalized images. Right column: histograms of the images in the center column.

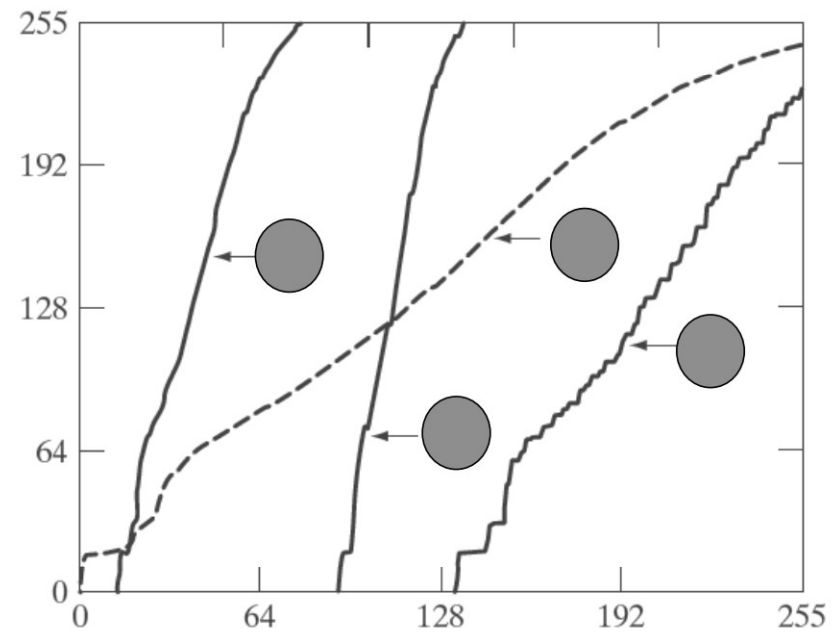


FIGURE 3.21 Transformation functions for histogram equalization. Transformations (1) through (4) were obtained from the histograms of the images (from top to bottom) in the left column of Fig. 3.20 using Eq. (3.3-8).

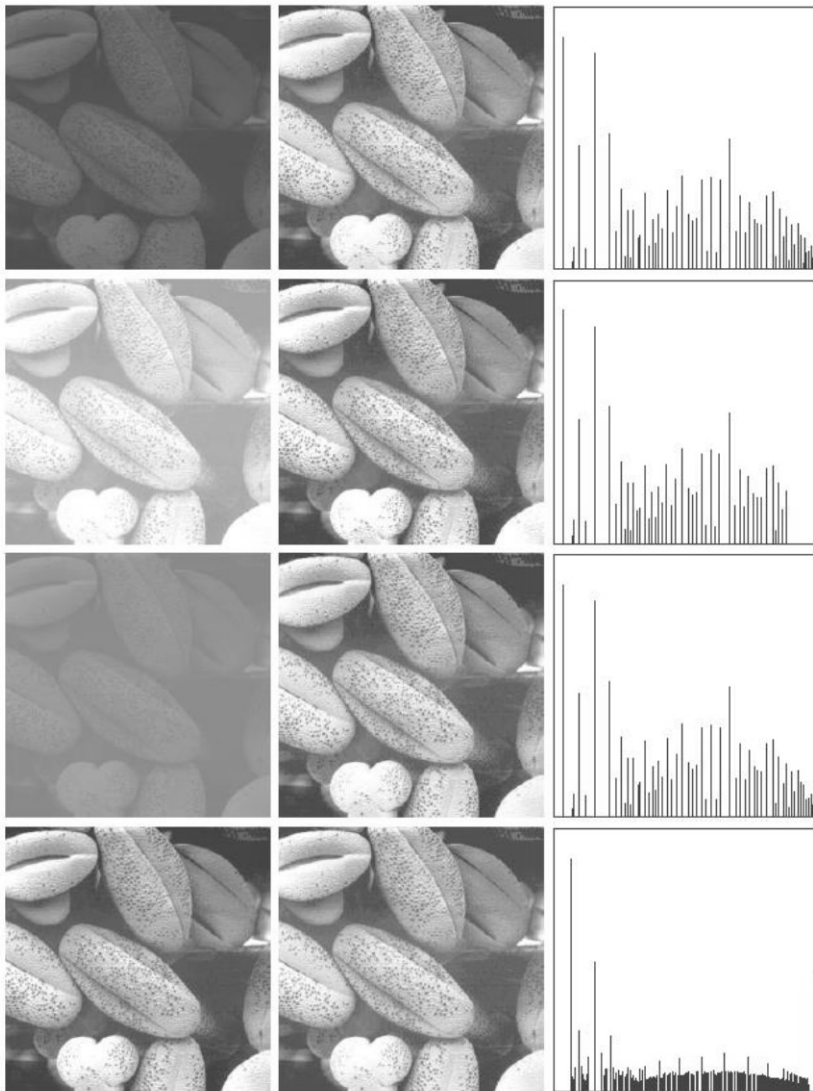


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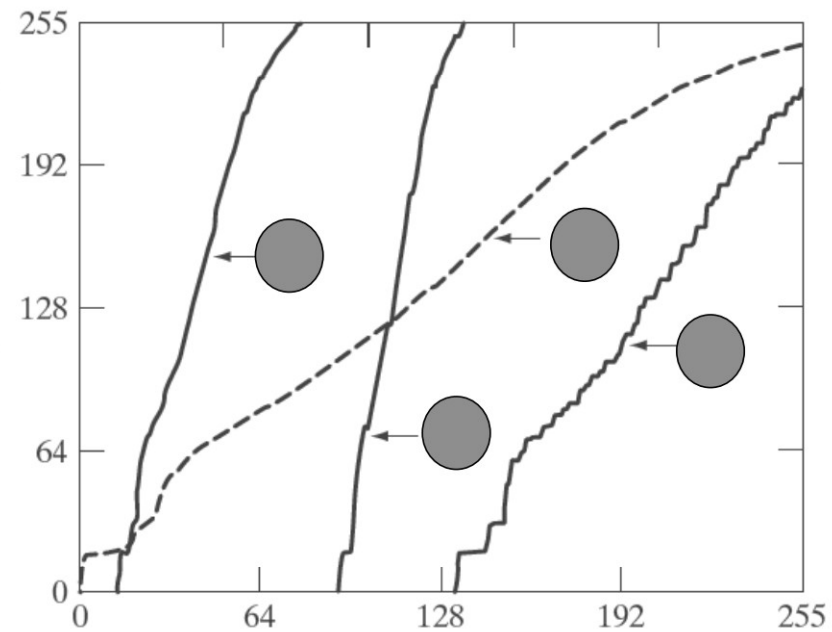


FIGURE 3.21 Transformation functions for histogram equalization. Transformations (1) through (4) were obtained from the histograms of the images (from top to bottom) in the left column of Fig. 3.20 using Eq. (3.3-8).

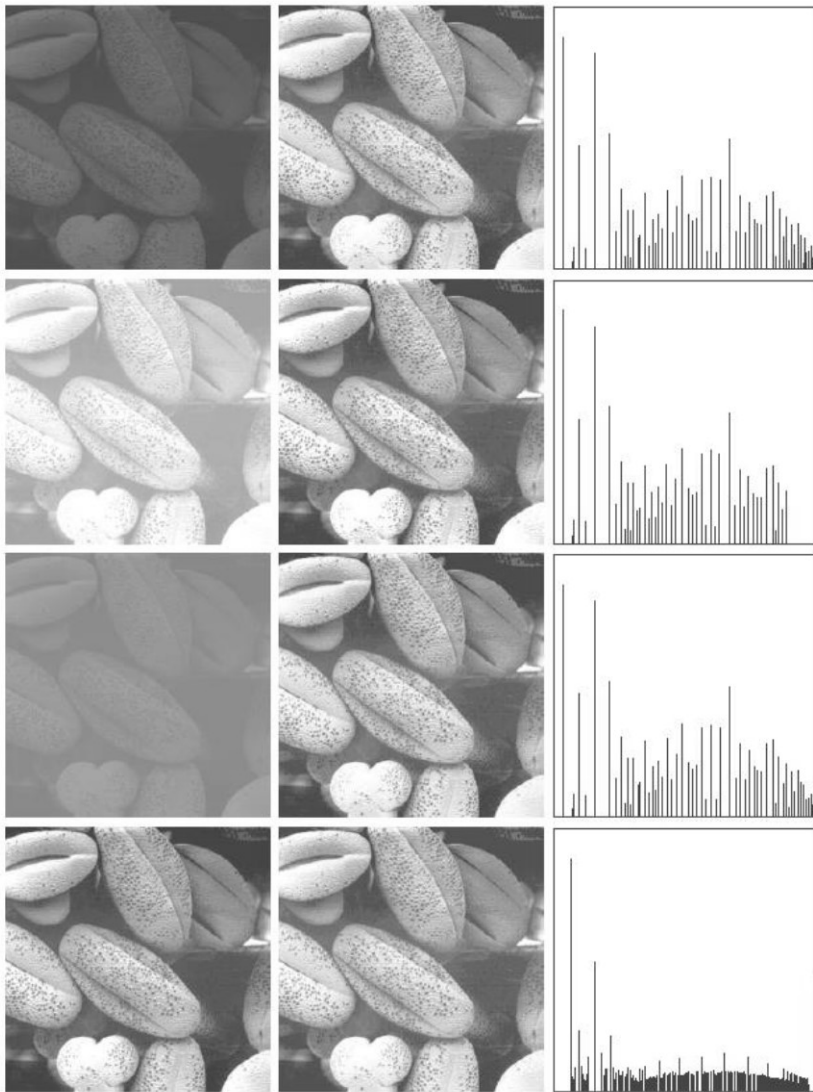


FIGURE 3.20 Left column: images from Fig. 3.16. Center column: corresponding histogram-equalized images. Right column: histograms of the images in the center column.

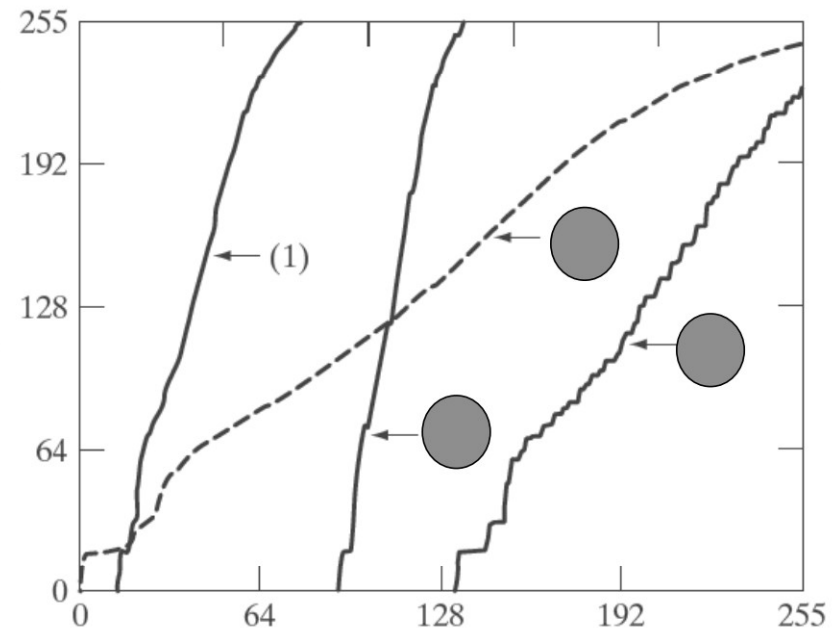


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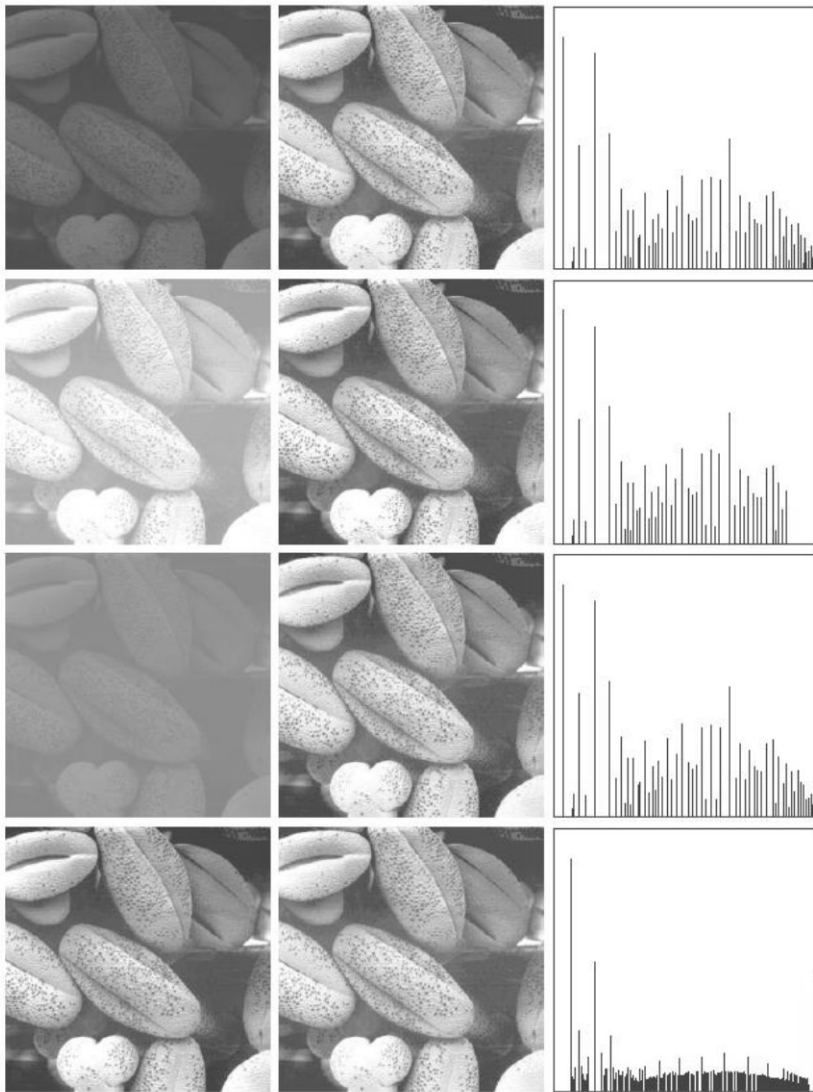


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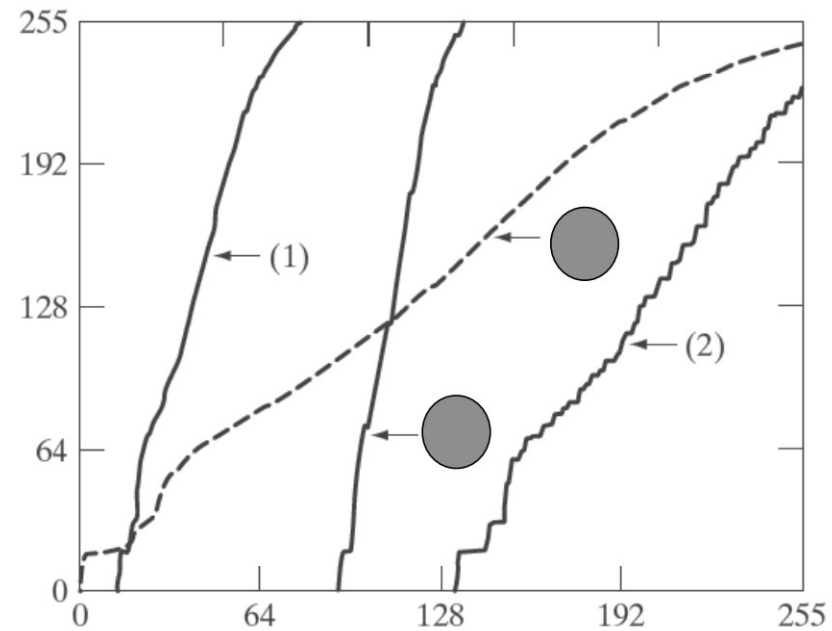


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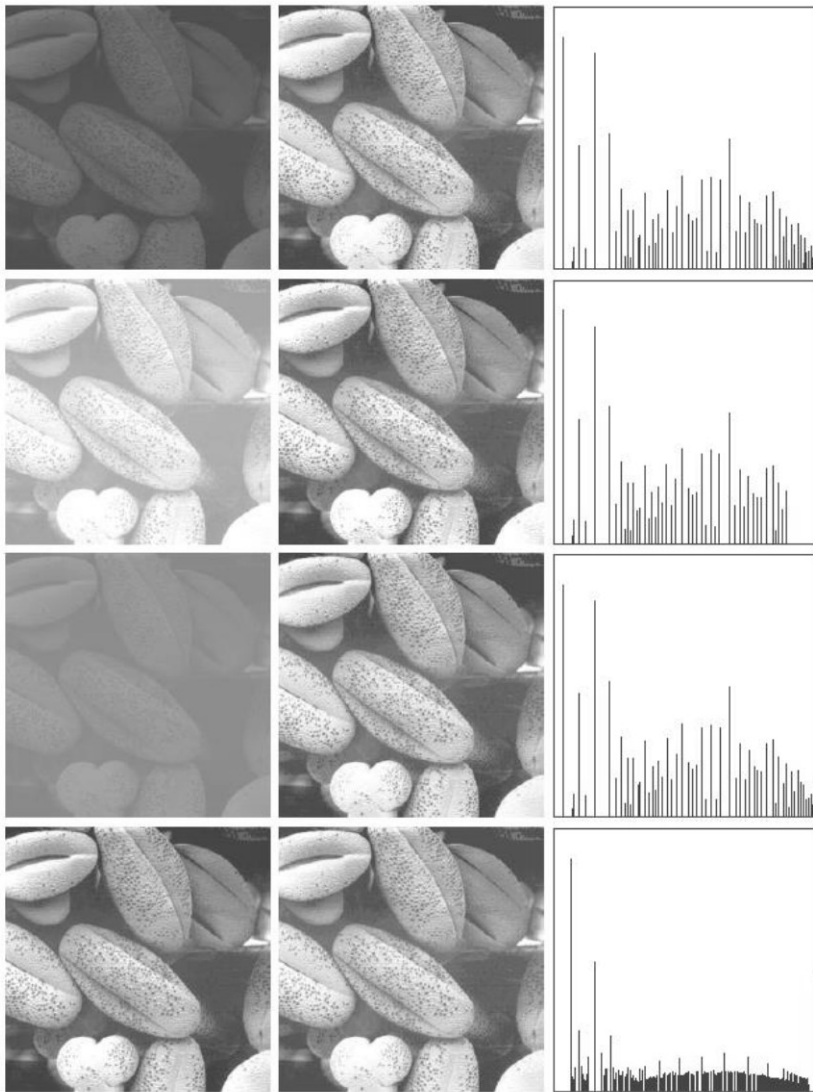


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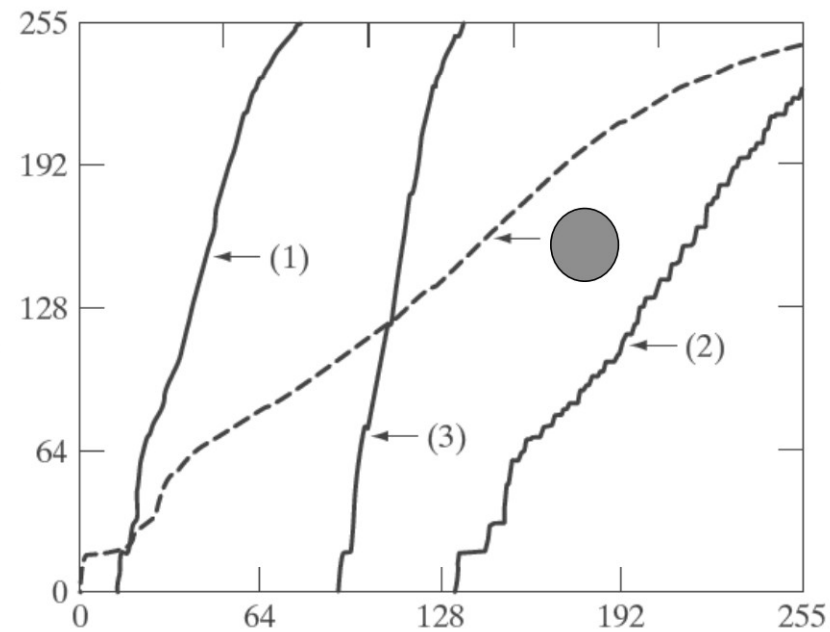


FIGURE 3.21 Transformation functions for histogram equalization. Transformations (1) through (4) were obtained from the histograms of the images (from top to bottom) in the left column of Fig. 3.20 using Eq. (3.3-8).

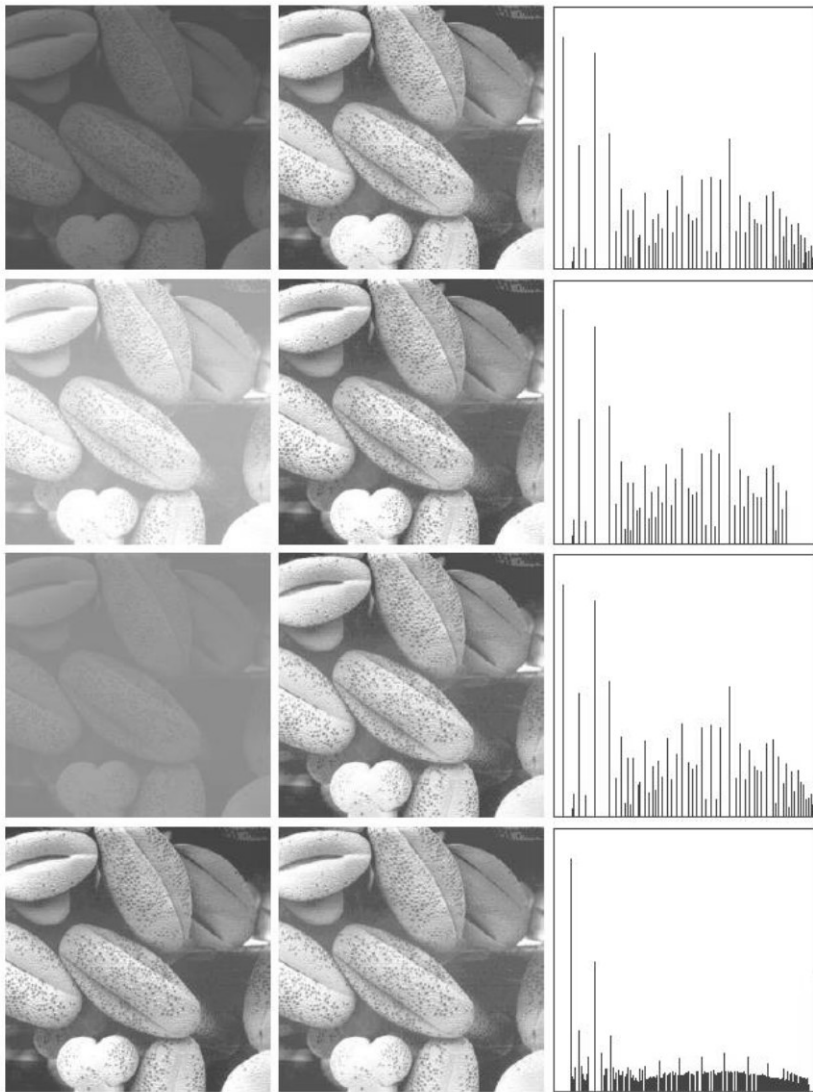


FIGURE 3.20 Left column: images from Fig. 3.16. Center column: corresponding histogram-equalized images. Right column: histograms of the images in the center column.

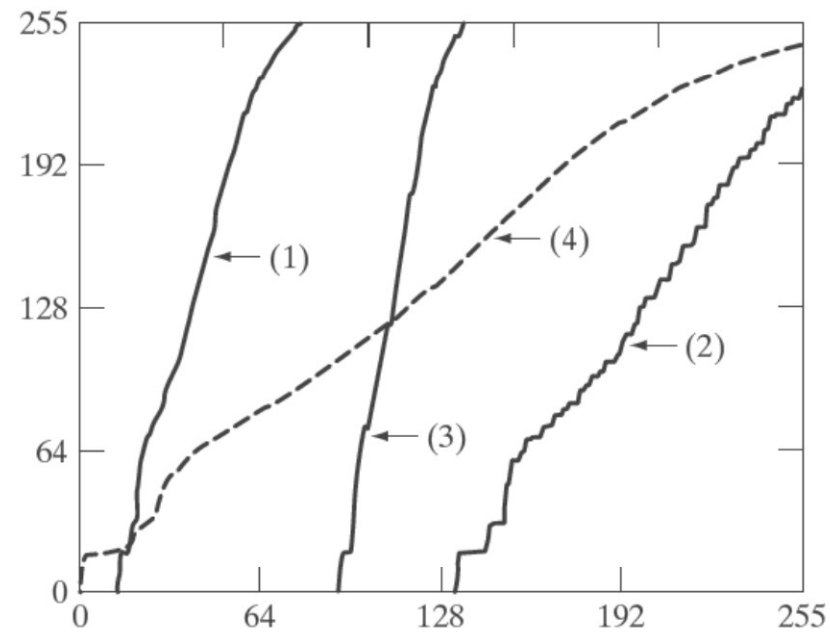
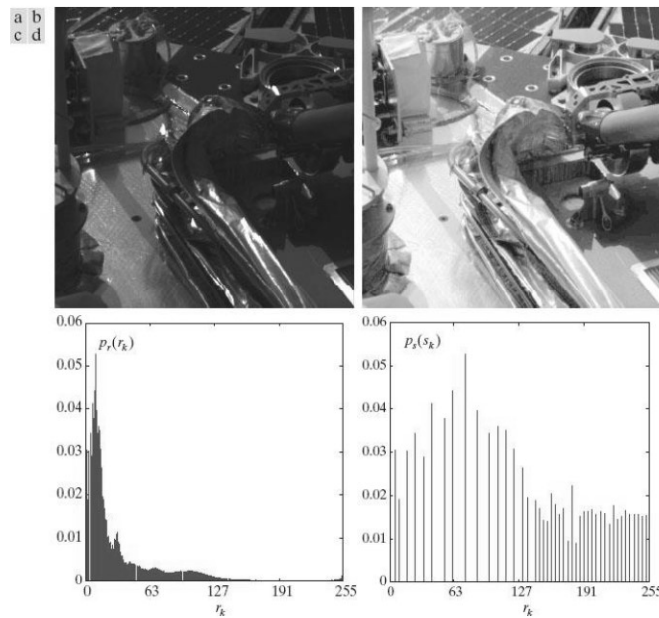


FIGURE 3.21 Transformation functions for histogram equalization. Transformations (1) through (4) were obtained from the histograms of the images (from top to bottom) in the left column of Fig. 3.20 using Eq. (3.3-8).

Figure 3.22

(a) Image from Phoenix Lander. (b) Result of histogram equalization. (c) Histogram of image (a). (d) Histogram of image (b). (Original image courtesy of NASA.)



Question

Question

Is histogram equalization always good?

No

Ques. Suppose that the intensity values in an image has PDF:

$$P_r(r) = \begin{cases} \frac{6r+2}{3(L-1)^2 + 2(L-1)} & \text{for } 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

Find the transformation fn $s = T(r)$ obtained through HE in continuous variables. Also show that $T(r)$ has uniform PDF.

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Find the transformation fn $s = T(r)$ obtained through HE in continuous variables. Also show that $T(r)$ has uniform PDF.

Sol. $S = T(r) = (L-1) \int_0^r P_r(w) dw$ $\frac{ds}{dr} = \frac{d}{dr} \left[\frac{3r^2+2r}{3(L-1)+2} \right]$

$= (L-1) \int_0^r \frac{6w+2}{3(L-1)^2+2(L-1)} dw$ $= \frac{6r+2}{3(L-1)+2} \quad \text{--- (1)}$

$= \frac{1}{3(L-1)+2} \left[\frac{6}{2} [w^2]_0^r + 2[w]_0^r \right]$ we know $P_s(s) = P_r(r) \times \frac{dr}{ds}$ put (1)

$= \frac{3r^2+2r}{3(L-1)+2}$ $P_s(s)$

$= \frac{6r+2}{3(L-1)^2+2(L-1)} \times \frac{3(L-1)+2}{6r+2}$

$= \frac{1}{L-1}$

$\therefore$ we show that $T(r)$ i.e s has uniform PDF

